

SECTION-I
Straight Objective Type

This section contains **20 multiple choice** questions. Each question has 4 choices (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** is correct.

- If $f(x) = 2 \tan^{-1} x + \sin^{-1} \left(\frac{2x}{1+x^2} \right)$ then for $x > 1$, $f(x) =$
 (1) $\sec^{-1}x$ (2) $4\sin^{-1}x$ (3) π (4) $\frac{\pi}{2}$
- If $\cos^{-1} \left(\frac{-\sqrt{3}}{2} \right) \sin^{-1} x = \frac{5\pi^2}{36}$ then which of the following is true?
 (1) $0 < x < \frac{1}{\sqrt{2}}$ (2) $x = \frac{1}{2} \sqrt{2 - \sqrt{\frac{3}{2}} + \frac{1}{\sqrt{2}}}$
 (3) $x > \frac{1}{2}$ (4) $\sqrt{2}x = \sqrt{1 + \sin \frac{\pi}{12}}$
- Range of function $f(x) = \cos^{-1}(-\{x\})$ where $\{x\}$ denotes fractional part of x , is
 (1) $\left(\frac{\pi}{2}, \pi \right)$ (2) $\left[\frac{\pi}{2}, \pi \right]$ (3) $\left[\frac{\pi}{2}, \pi \right)$ (4) $\left(0, \frac{\pi}{2} \right]$
- Solution of $3(\cot^{-1} x)^2 - 4\sqrt{3} \cot^{-1} x + 3 > 0$ is
 (1) $\left(\frac{\pi}{6}, \frac{\pi}{3} \right)$ (2) $\left(-\infty, \frac{\pi}{6} \right) \cup \left(\frac{\pi}{3}, \infty \right)$
 (3) $\left(-\infty, \cot \sqrt{3} \right) \cup \left(\cot \frac{1}{\sqrt{3}}, \infty \right)$ (4) $(0, \pi)$
- $\sum_{n=1}^{10} \tan(\tan^{-1}(2^n - 2)) =$
 (1) 2046 (2) 2026 (3) 2016 (4) 2006
- Which of the following functions pair is identical ?
 (1) $f(x) = x, g(x) = \frac{1}{1/x}$
 (2) $f(x) = \cos x, g(x) = \frac{1}{\sqrt{1 + \tan^2 x}}$
 (3) $f(x) = \tan^{-1}x + \tan^{-1} \frac{1}{x}, g(x) = \sin^{-1}x + \cos^{-1}x$
 (4) $f(x) = \cos(\sin^{-1}x), g(x) = \sqrt{1 - x^2}$

7. For $x \in (0,1)$, let $\alpha = \sin^{-1}x$, $\beta = x$, $\gamma = \tan^{-1}x$, $\delta = \cot^{-1}x - \frac{\pi}{2}$. Which of the following is true?
- (1) $\delta > \alpha > \beta > \gamma$ (2) $\beta > \alpha > \gamma > \delta$ (3) $\alpha > \beta > \gamma > \delta$ (4) $\beta > \alpha > \delta > \gamma$
8. The value of $\sin \left(\frac{1}{4} \sin^{-1} \frac{\sqrt{63}}{8} \right)$ is
- (1) $\frac{1}{2}$ (2) $\frac{1}{3}$ (3) $\frac{1}{2\sqrt{2}}$ (4) $\frac{1}{5}$
9. The value of $\sec \left[\sin^{-1} \left(-\sin \frac{50\pi}{9} \right) + \cos^{-1} \left(\cos \frac{31\pi}{9} \right) \right]$ is
- (1) $\sec \left(\frac{10\pi}{9} \right)$ (2) $\sec \left(\frac{\pi}{9} \right)$ (3) 1 (4) -1
10. $2\cot^{-1}(7) + \cos^{-1} \left(\frac{3}{5} \right)$, in principal value, is equal to
- (1) $\tan^{-1} \left(\frac{41}{117} \right)$ (2) $\tan^{-1} \left(\frac{117}{125} \right)$ (3) $\cos^{-1} \left(\frac{44}{125} \right)$ (4) $\cos^{-1} \left(\frac{44}{117} \right)$
11. The solution set of the inequality $\tan^{-1}x + \sin^{-1}x \geq \frac{\pi}{2}$ is
- (1) $[-1, 1]$ (2) $\left[\sqrt{\frac{\sqrt{5}-1}{4}}, 1 \right]$ (3) $\left[\sqrt{\frac{\sqrt{5}-1}{2}}, 1 \right]$ (4) $\left[\frac{\sqrt{5}-1}{2}, 1 \right]$
12. The algebraic expression for $\tan \left(\sin^{-1} \cos \tan^{-1} \frac{x}{2} \right)$ is
- (1) $\frac{2}{x}$ (2) $\frac{x}{2}$ (3) $\frac{1}{x}$ (4) $\frac{2}{|x|}$
13. Expression $\sum_{n=1}^{\infty} \cot^{-1}(n^2 - 3n + 3)$ simplifies to
- (1) $\frac{\pi}{4}$ (2) $\frac{\pi}{2}$ (3) $\frac{3\pi}{4}$ (4) π
14. If $\sin^{-1} x + \sin^{-1} y + \sin^{-1} z = \frac{3\pi}{2}$ and $f(1) = 2$, $f(p+q) = f(p) \cdot f(q) \forall p, q \in \mathbb{R}$, then $x^{f(1)} + y^{f(2)} + z^{f(3)} - \frac{x+y+z}{x^{f(1)} + y^{f(2)} + z^{f(3)}} =$
- (1) 0 (2) 1 (3) 2 (4) 3
15. If $A = 2 \tan^{-1}(2\sqrt{2} - 1)$ and $B = 3 \sin^{-1} \left(\frac{1}{3} \right) + \sin^{-1} \left(\frac{3}{5} \right)$, then -
- (1) $A = B$ (2) $A < B$ (3) $A > B$ (4) None of these
16. Which of the following is negative
- (1) $\cos(\tan^{-1}(\tan 4))$ (2) $\sin(\cot^{-1}(\cot 4))$
(3) $\tan(\cos^{-1}(\cos 5))$ (4) $\cot(\sin^{-1}(\sin 4))$

17. The number of solutions of the equation $\tan^{-1}(4\{x\}) + \cot^{-1}(x + [x]) = \frac{\pi}{2}$, is (where $[.]$ denotes greatest integer function and $\{.\}$ fractional part function)
- (1) 2 (2) 3 (3) 1 (4) 4
18. Number of integers in domain of $f(x) = \log \left(\tan^{-1} \left(\frac{x^3 - 6x^2 + 11x - 6}{x(e^x - 1)} \right) \right) + \sqrt{10 - x}$ are
- (1) 5 (2) 6 (3) 8 (4) 7
19. Range of $\sin^{-1} \left[x^2 + \frac{1}{2} \right] + \cos^{-1} \left[x^2 - \frac{1}{2} \right]$ where $[.]$ denotes the greatest integer function, is -
- (1) $\left\{ \frac{\pi}{2}, \pi \right\}$ (2) $\{\pi\}$ (3) $\left\{ \frac{\pi}{2} \right\}$ (4) None of these
20. If $[x]$ denotes the integral part of x and $k = \sin^{-1} \frac{1+t^2}{2t} > 0$, then the integral value of α for which the equation $(x - [k])(x + \alpha) - 1 = 0$ has integral root is -
- (1) -1 (2) 1 (3) 2 (4) None of these

SECTION-II

(Integer Type Questions)

This section contains **Five (05)** questions. The answer to each question is **NUMERICAL VALUE** with two digit integer and decimal upto one digit

21. If $1 \leq x \leq y \leq z$ and $\sin^{-1} x + \sec^{-1} y - \cot^{-1} z = \frac{\pi}{4}$ then $x^2 - yz + z^2 =$
22. The number of roots of the equation $\sin^{-1}(1-x) + 2\sin^{-1} x = \frac{\pi}{2}$, is
23. If $\sum_{\alpha=1}^{\infty} \operatorname{cosec}^{-1} \sqrt{1+4\alpha^4} = \frac{a\pi}{b}$ (where a, b are coprime number) then $(a + b)$ is equal to
24. If $\sin^{-1}(\sin 5) > x^2 - 4x$, then the number of possible integral values of x is
25. If $\sin^{-1}(\sin^{-1} 1/5 + \cos^{-1} x) = 1$ then $5x$ equals

Answer Key_DPP_04

1.	(3)	2.	(1)	3.	(3)	4.	(3)	5.	(2)	6.	(4)	7.	(3)
8.	(3)	9.	(4)	10.	(3)	11.	(3)	12.	(4)	13.	(3)	14.	(3)
15.	(3)	16.	(4)	17.	(1)	18.	(4)	19.	(2)	20.	(1)	21.	(1)
22.	(2)	23.	(5)	24.	(3)	25.	(1)						